92. The technical details can be found in Carlo Rovelli, “Is Time’s Arrow Perspectival?” in *The Philosophy of Cosmology*, eds. K. Chamcham et al. (Cambridge, UK: Cambridge University Press, 2017), https://arxiv.org/abs/1505.01125.

93. In the classical formulation of thermodynamics, we describe a system by specifying in the first place some variables on which we assume we can act from the outside (moving a piston, for example), or which we assume we can measure (a relative concentration of components, for xample). These are the “thermodynamic variables.” Thermodynamics is not a true description of the system; it is a description of *these* variables of the system—those through which we assume we are able to interact with the system.

94. For example, the entropy of the air in this room has a value taken from air as a homogeneous gas, but it changes (diminishes) if I measure its chemical composition.

95. A contemporary philosopher who has shed light on these aspects of the perspectival nature of the world is Jenann T. Ismael, *The Situated Self* (New York: Oxford University Press, 2007). Ismael has also written an excellent book on free will: *How Physics Makes Us Free* (New York: Oxford University Press, 2016).

96. David Z. Albert, in *Time and Chance* (Cambridge, MA: Harvard University Press, 2000), proposes to elevate this fact to a natural law, and calls it “past hypothesis.”

11. WHAT EMERGES FROM A PARTICULARITY

97. This is another common source of confusion, because a condensed cloud seems more “ordered” than a dispersed one. It isn’t, because the speed of the molecules of a dispersed cloud are all small (in an ordered manner), while, when the cloud condenses, the speeds of the molecules increase and spread in phase space. The molecules concentrate in physical space but disperse in phase space, which is the relevant one.

98. See, in particular, Stuart A. Kauffman, *Humanity in a Creative Universe* (New York: Oxford University Press, 2016).

99. The importance of the existence of this ramified structure of interactions in the universe for the understanding of the growth of local entropy is discussed, for instance, by Hans Reichenbach, in *The Direction of Time* (Berkeley: University of California Press, 1956). Reichenbach’s text is fundamental for whoever has doubts about these arguments or is interested in pursuing them in more depth.

100. On the precise relation between traces and entropy, see Reichenbach, *The Direction of Time*, in particular the discussion on the relation among entropy, traces, and *common cause*, and Albert, *Time and Chance*. A recent approach can be found in D. H. Wolpert, “Memory Systems, Computation and the Second Law of Thermodynamics,” *International Journal of Theoretical Physics* 31 (1992): 743–85.

101. On the difficult question of what “cause” means to us, see Nancy Cartwright, *Hunting Causes and Using Them: Approaches in Philosophy and Economics* (Cambridge, MA: Cambridge University Press, 2007).

102. “Common cause,” in Reichenbach’s terminology.

103. Bertrand Russell, “On the Notion of Cause,” *Proceedings of the Aristotelian Society*, N. S. 13 (1912–1913): 1–26.

104. Cartwright, *Hunting Causes and Using Them*.